Script Tools

This document contains a list of the functions and features available in the software suite for the Lab's Drawing set.

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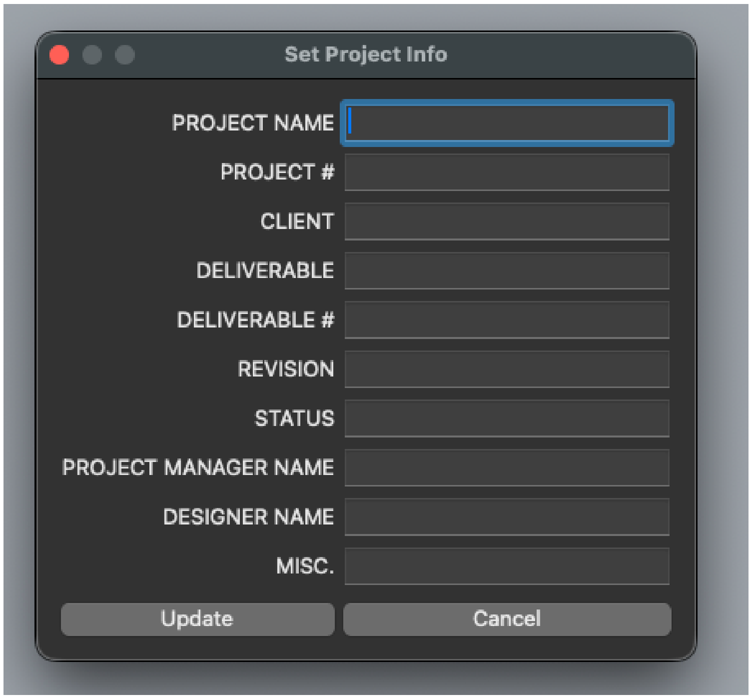
ProjectInfo

Script name is **ProjectInfo.py**

Sets global properties for the project in the Rhino document that are referenced by the layout block instance. This allows the user to set the parameters about the project they are working on without having to go into the document user text window.

These parameters include

- PROJECT NAME - Name of the project on the estimate
- PROJECT # - Assigned Project Number
- CLIENT - Client name
- DELIVERABLE - The name of the deliverable
- DELIVERABLE # - Number of the deliverable assigned via our estimate
- REVISION - Major revision number, should be updated with each submission to the client (R1, R2, etc.)
- STATUS - Project Status. Client Review, Engineering Review, Ready for Production, etc.
- PROJECT MANAGER NAME - Self explanatory
- DESIGNER NAME - You
- MISC. - Any relevant info that doesn't fit in another category



When you press the "Update" button, it will store these parameters in the document and update any existing layouts you already have.

Assembly Manager

For more info on how this works, see this article.

The Assembly Manager is the basis for utilizing the information modeling system that we are developing at The Lab. This application is designed to expedite the process of organizing, categorizing, labeling, and prepping geometry for production.

The basic premise of the assembly manager is to break down a complex structure into a simple hierarchy.

1. Assembly - An assembly is composed of a set of components.
2. Component - A component is composed of parts. A component represents an object that one or two people could build together as well as a unit that could be broken down for shipping.
3. Part - A part is the most basic building block for something that needs to be fabricated. Imagine this as a CNC cut part or a simple object that could be made by hand.

Given a simple set of rules, the Assembly manager will automatically sort and label components and parts for you into an assembly.

Currently the assembly manager provides two major functions. First, there is the create assembly function, which will

1. Prompt you to select a set of parts (it only currently accepts polysurfaces)
2. Categorize all the different parts you have selected based on their specific geometry.
3. For each group that the parts belong to it will create a component.
4. Then it compares the components to one-another to find out if they are equivalent based on the quantity of a set of parts.
5. Lastly, it consolidates components down to a set of layers.

Then there is the "**Lay Parts Flat**" function, which will

1. Isolate one of each part in the assembly and orient it to lay flat on the construction plane
2. Create a text object below it that labels the part, tells you the quantity of the parts in the assembly, and the material thickness.

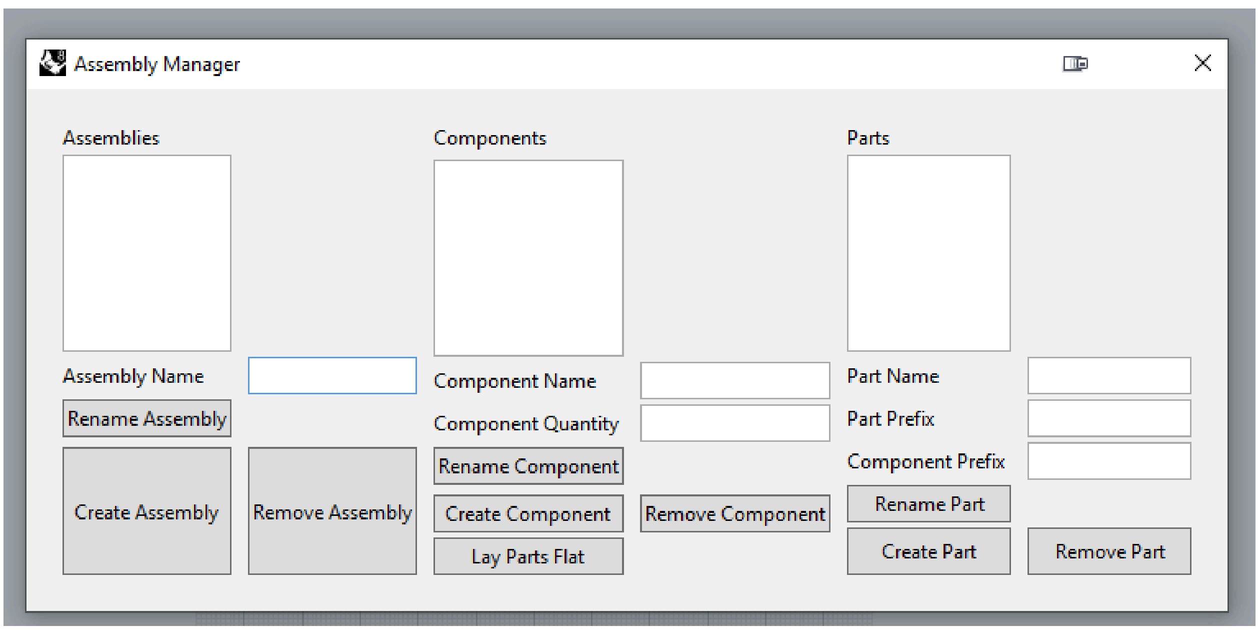
Prepping a Model for Assembly Manager

Before you feed a model into the assembly manager, you will want to prep your model so that the system knows how to deal with your parts.

1. Clean up any excess geometry out of your model. Do not feed it anything that you do not want to categorize. This will cause confusion in terms of material quantities as well as make the process of sorting through all the geometry take much longer than necessary.
2. Decide what parts should be represented as a component and group them together. These groups should **NOT BE NESTED** . If you nest groups the system won't know which group is the parent and which one is the child. To make this grouping process faster, there is a **regroup** function, which will dissolve all groups in a set of selected objects and regroup them into a single group.

Create an Assembly

After you have your model prepped, you can run the **AssemblyManager** command which will bring up this window.

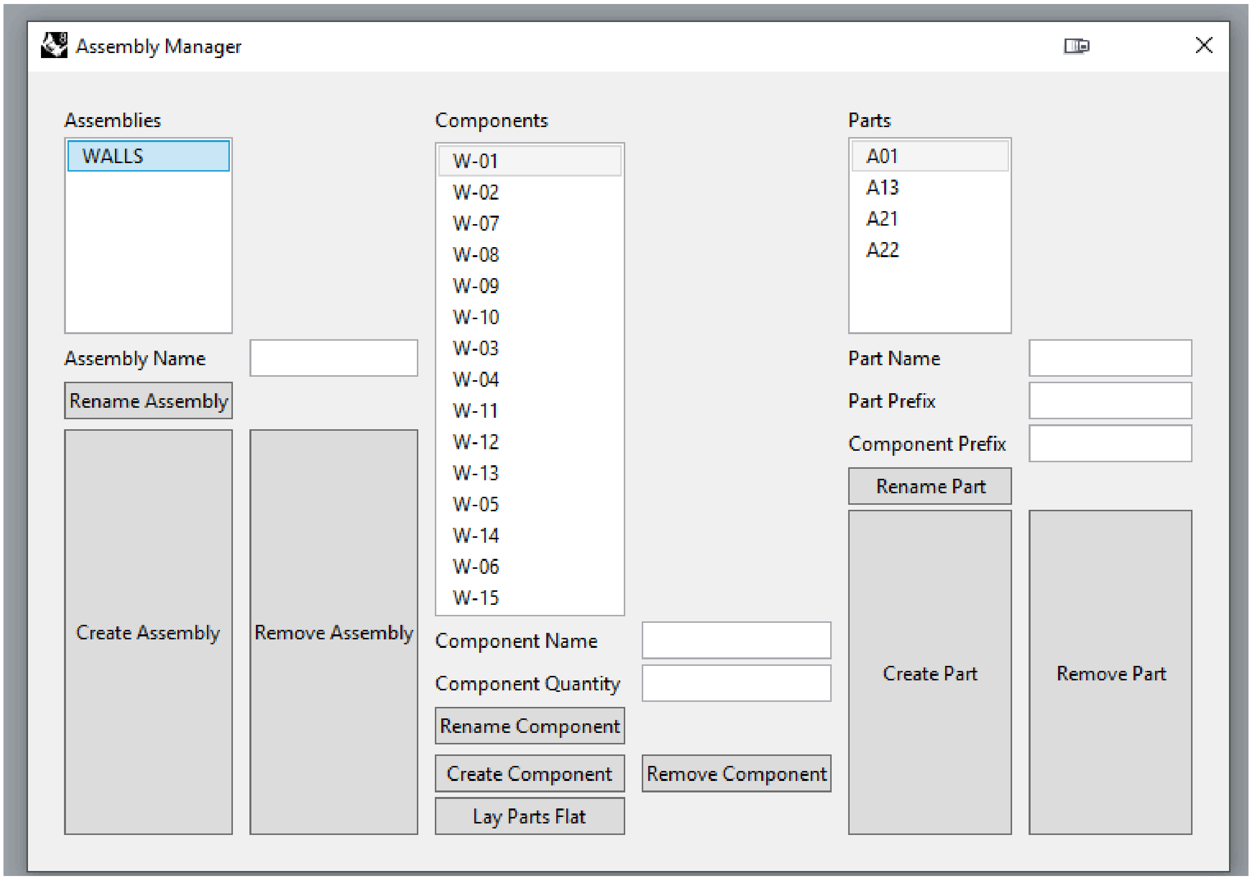


To create an assembly, you first need to provide a name to the assembly by filling in the "Assembly name" text box. Then, you should ideally add in a "Part Prefix" and "Component Prefix". This will apply a prefix to each numbered component and part. This will help with identifying parts when auto-labeling functions are available.

Once you have filled in those fields you can press the **create assembly** button and it will prompt you to select the geometry that you want to use to create the assembly.

This will now go through the process of categorizing all the parts and components in the system and transfer those newly categorized parts the the "SHOP" parent layer.

After it has completed the categorization process, the assembly manager window will pop back up and it will now have information in the list boxes labeled "Assemblies", "Components", and "Parts"



If you first double click on an assembly, then double click on a component in the list box, it will populate the fields associated with that component. It will tell you the quantity of the component in the assembly, and what parts are in it.

Lay Parts Flat

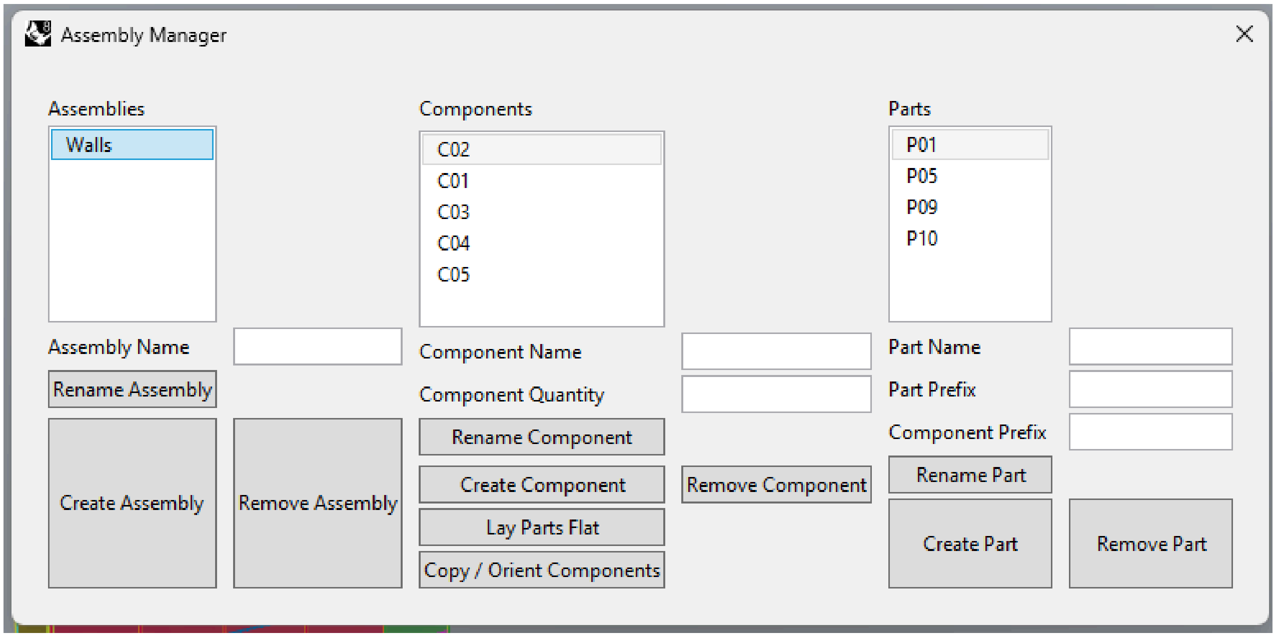
If you are ready to take all the parts of the assembly and lay them down on the construction plane for CNC, you can double click on the assembly you want to lay flat to select it, and then simply press the "**Lay Parts Flat**" button.

This will then take all the parts in the assembly and lay them out on the construction plane near the origin.

Currently you will still have to extract the important geometry for tool pathing, but more features are planned for the future to help with that process.

Copy / Orient Components

When you are ready to start creating drawings for the shop for each component, you will want to use the "Copy / Orient Components button"



This function will

1. Find all the components in an assembly
2. Create a new layer set under the "DRAWINGS" parent layer for the assembly, components, parts.
3. Copy one instance of each component to the created components.
4. Move each component to the X-axis and attempt to orient it along the X-Axis utilizing the **OrientToWorld** function.

The purpose of this function is to allow you to manipulate the components for your drawing sets without disturbing the structure of what is in the **SHOP** layer set. The **SHOP** layer set created by the assembly manager after categorizing parts needs to be preserved as to ensure that it can accurately count parts during the **lay parts flat** operation.

Regroup

This function was created specifically to help with the process of prepping a model for the assembly manager system. It simply takes in a set of geometry, dissolves all existing groups, and regroups them under a single group.

Split3pt

Split 3 point is a simple function that takes in a Brep and prompts the user to pick 3 point that define a plane to split the object with.

It then creates the plane, splits the selected objects with it, and deletes the plane.

OrientToWorld

This function rotates a set of selected objects automatically using gradient decent. The function is purely based on the volume of a bounding box oriented to the world coordinates.

The function was developed to be a part of the **Copy / Orient Parts** function in the assembly manager, but there is also a function the the user can call directly via the command line.

The function does not always work as expected, as the solution space may have multiple peaks and valleys, like any gradient descent solver.

LabelDetail

The **LabelDetail** function will automatically apply labels to parts and components in a detail view.

This function requires that you utilize the **assembly manager** to create a layer structure for the system to read from.

This function applies text dots to the Volume Centroid of every part visible in the detail viewport and sets the text of the text dot to the name of the part or component.

To use this function

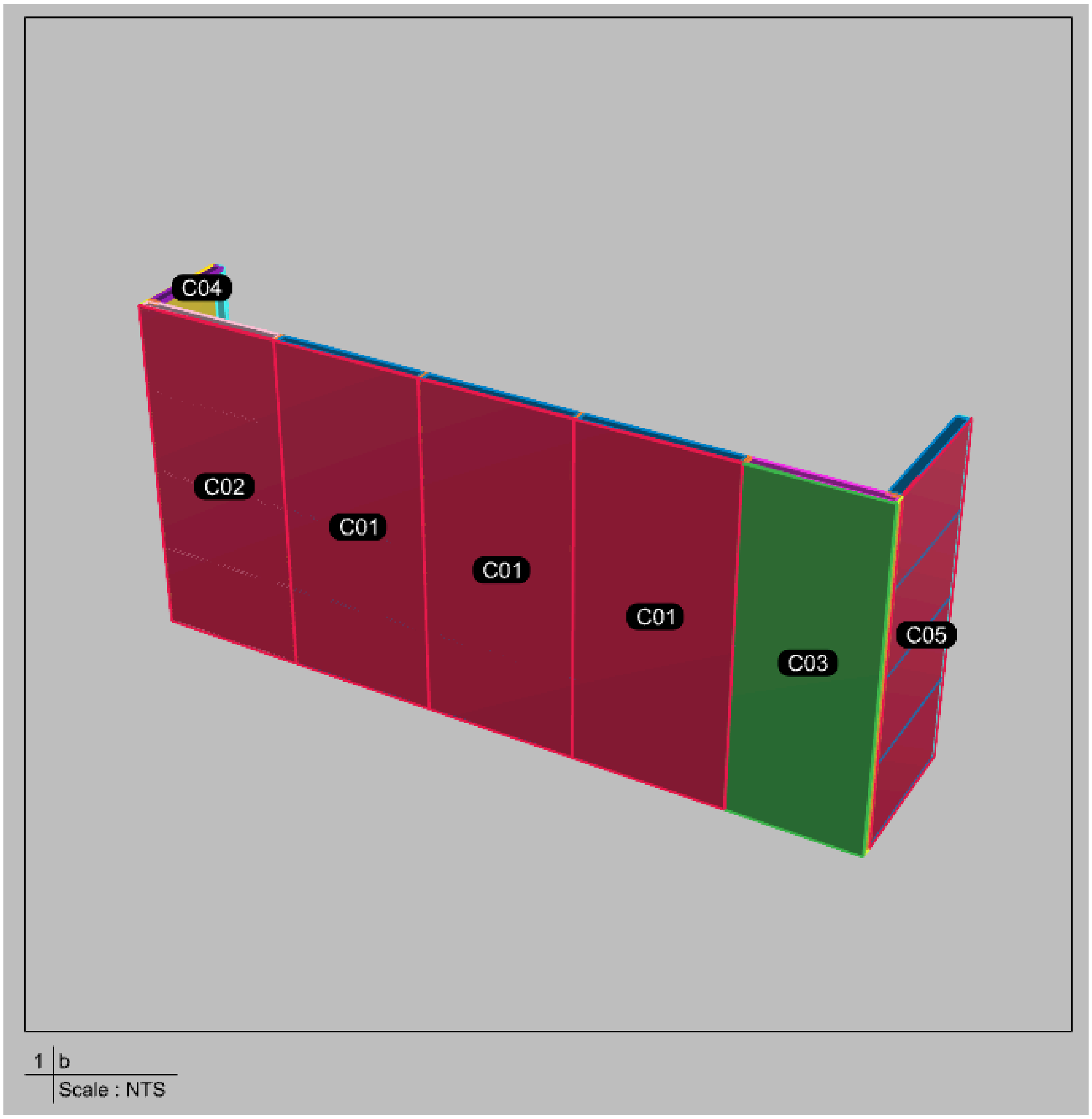
1. Make sure you have created an assembly with the assembly manager and used the "**Copy / Orient Parts**" function to create copies of your objects. These copies will be on the **DRAWINGS** parent layer, which is where this function will look for objects to label.
2. Create a detail view on a layout page with one of copied components within the view of the detail.
3. Run the **LabelDetail** function and select the Detail you want to label.

After you have selected a detail, the system will bring up a menu for you to select your label level. There are two levels to the labeling.

1. **Assembly** - The assembly level will label the components within an assembly. This will only produce labels for objects that are on the **SHOP** layer, which is created with the **Create Assembly** function on the **Assembly Manager**
2. **Component** - The component level will label the parts in the components. This will only produce labels for objects that are on the **DRAWINGS** layer which is created using the **Copy / Orient Parts** function in the **Layout Manager**

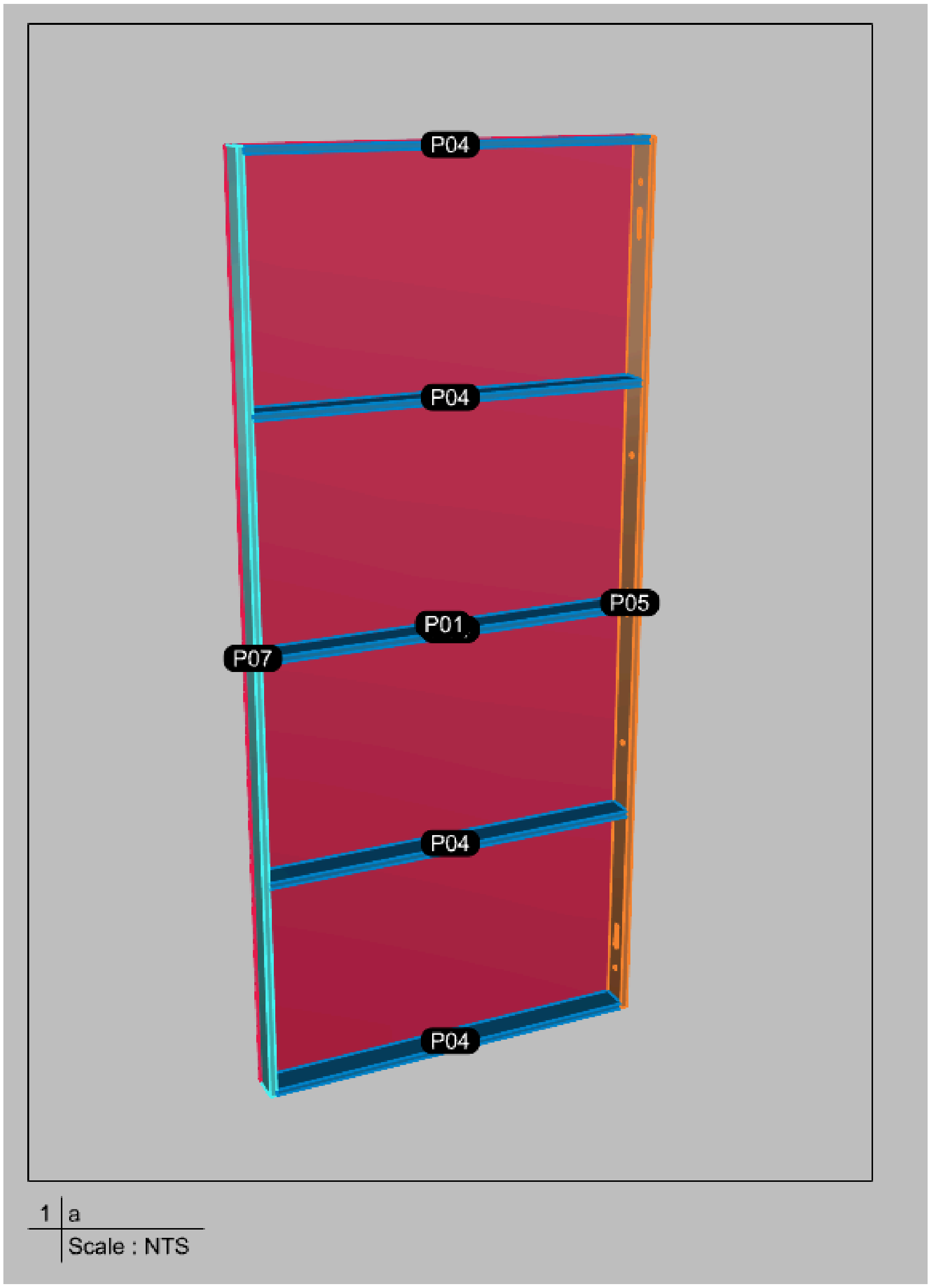
Assembly Level Labeling

If any components that are on the **SHOP** layer are visible in the viewport you selected, you will see the system apply text dots to the center of the components.



Component Level Labeling

If any parts that are on the **DRAWINGS** layer are visible in the viewport you should see the system apply text dots to the pages.

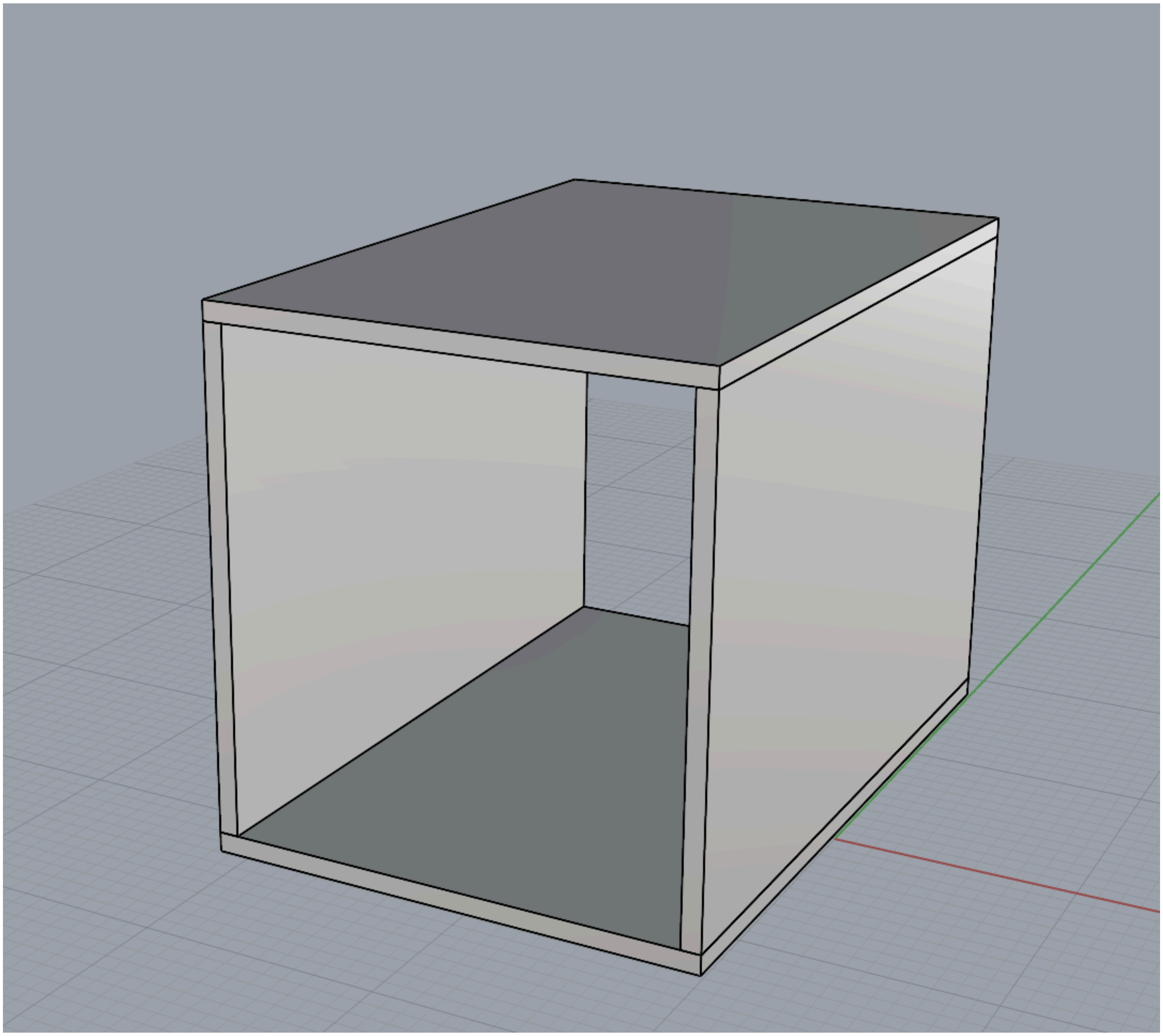


MotionTrace

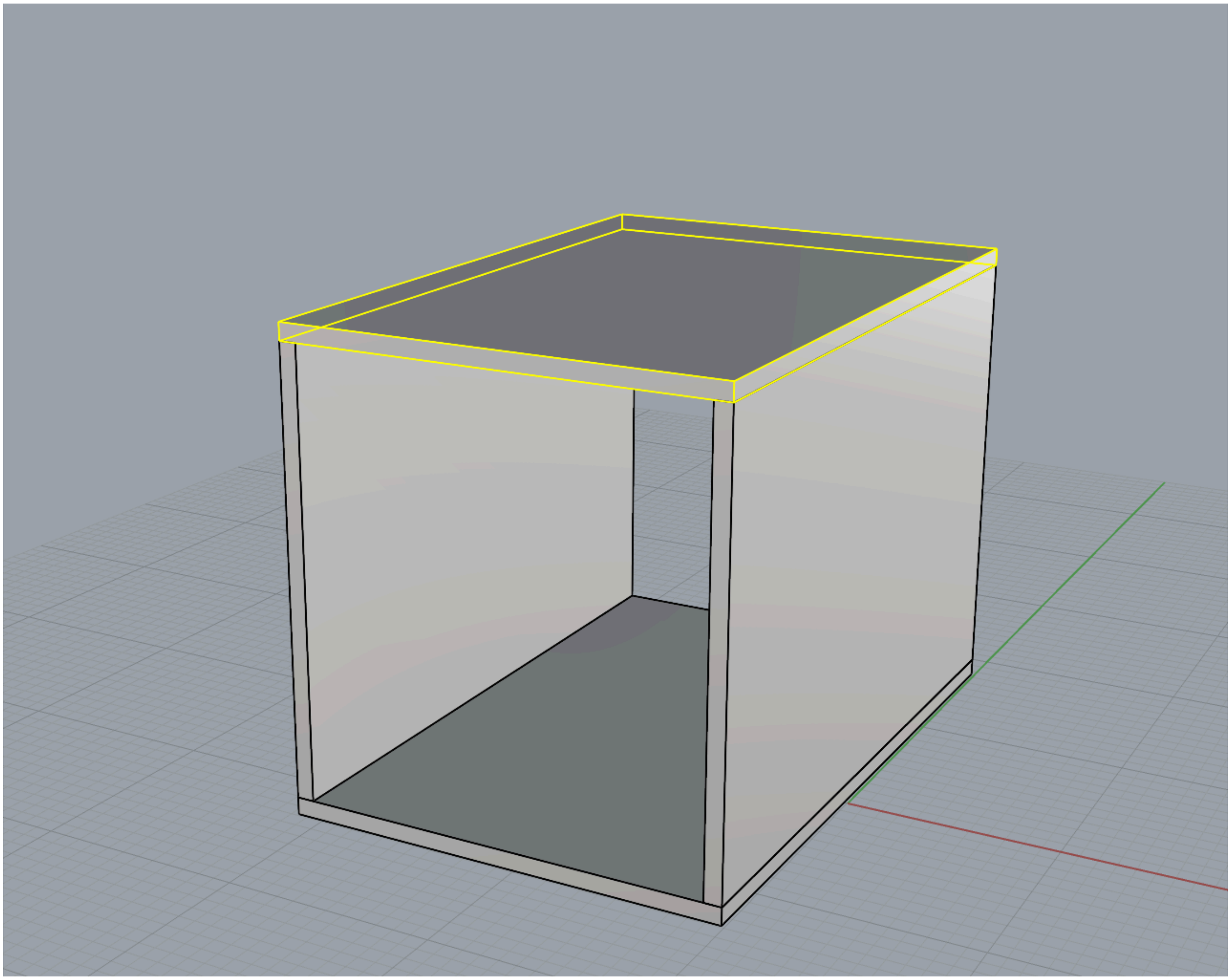
The MotionTrace function is a quick function for showing motion in details.

Essentially, it is a point to point move function that leaves behind the edges of the moved object as hidden linework on the annotations layer.

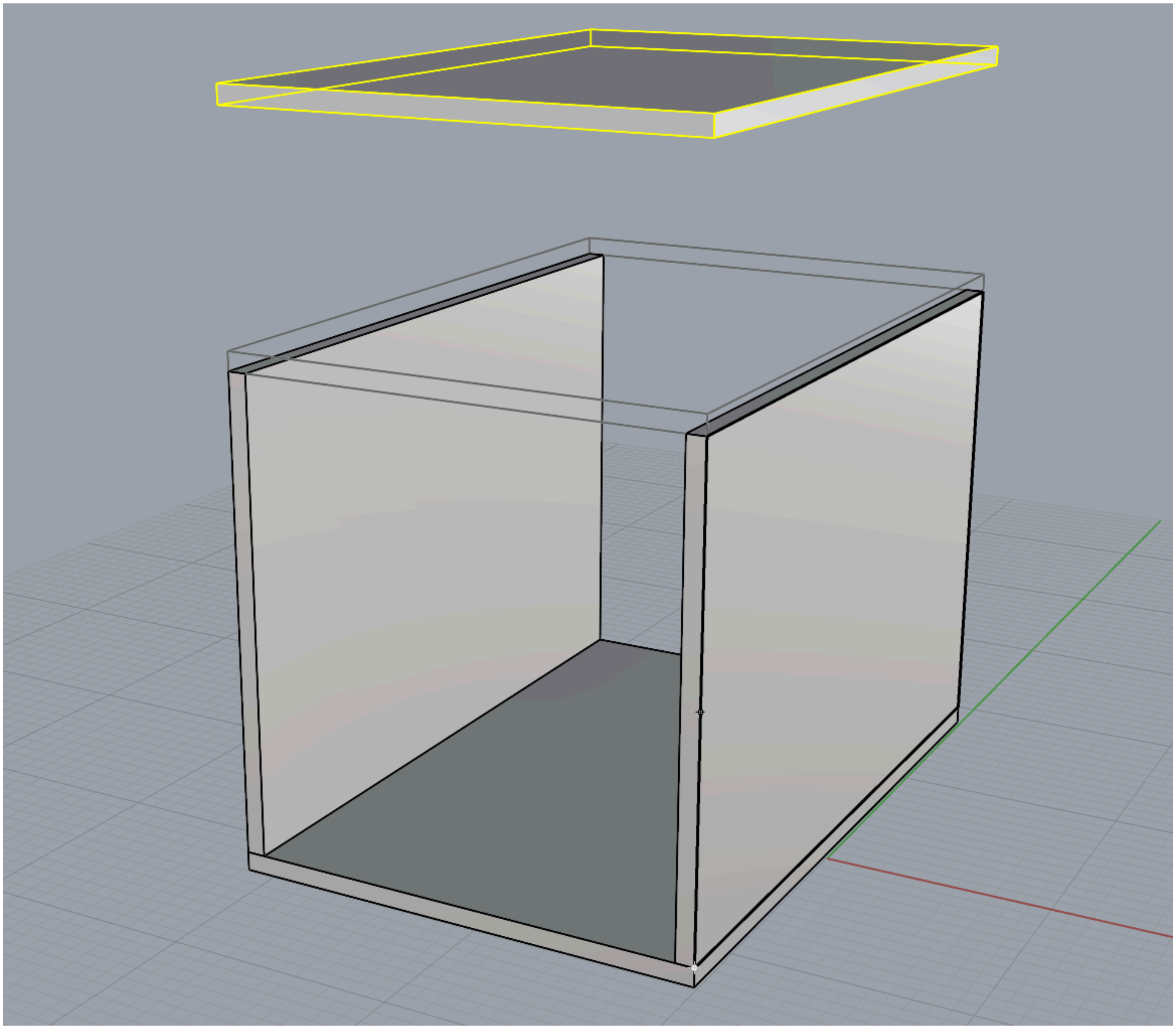
For example, you want to show the the assembly of the top of this box.



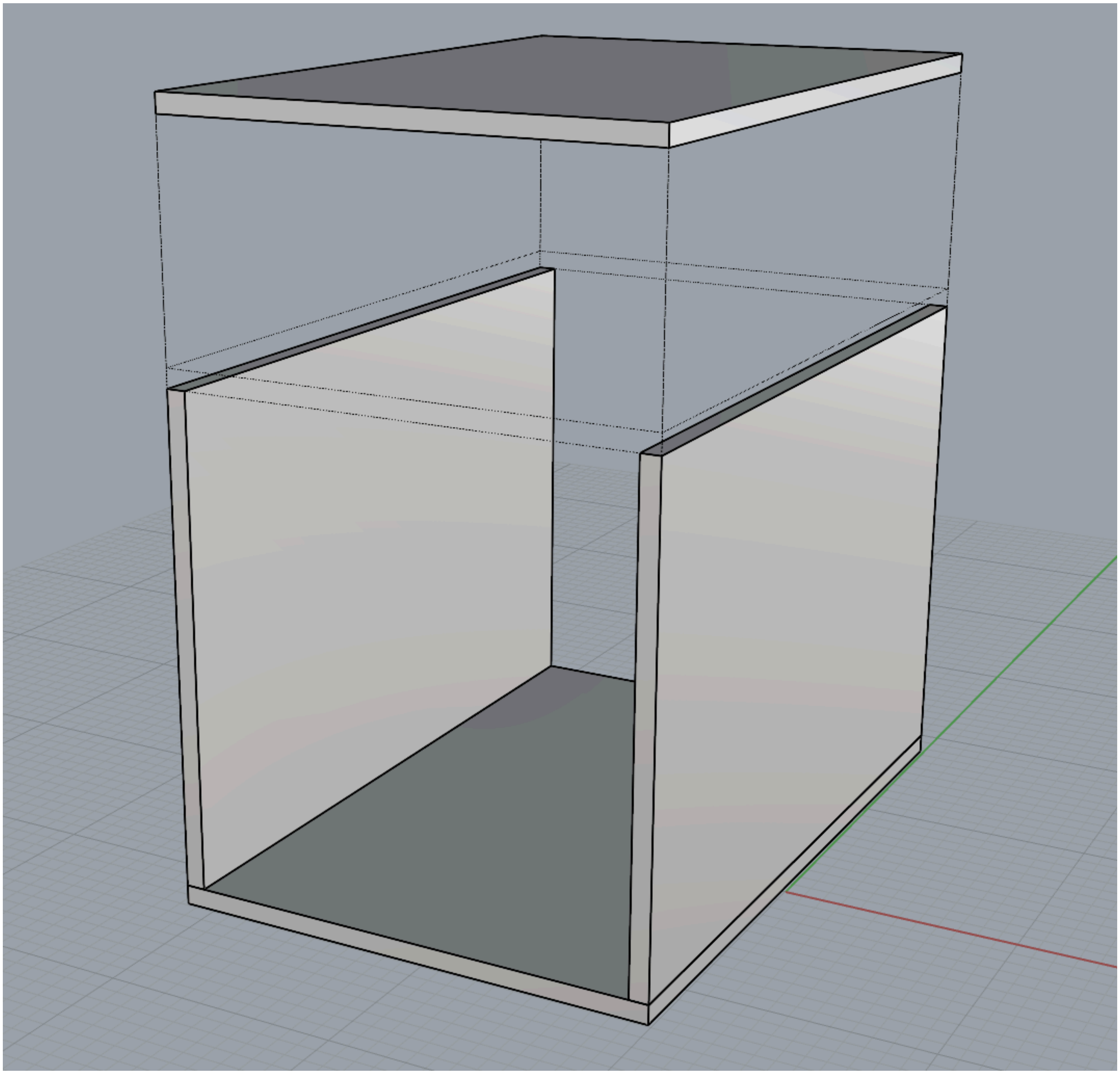
You start the MotionTrace function. It prompts you to select the objects you want to move.



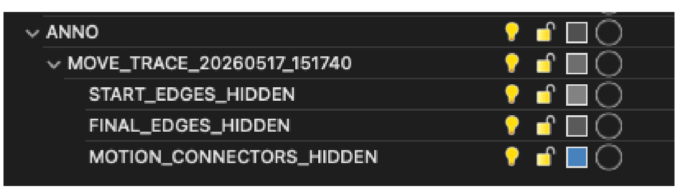
You select the start and end points of the motion.



And it leaves behind the edges of the object and lines connecting the two objects.



It should create a new layer "ANNO" if it doesn't already exist and it saves the motion trace on layers, such that you can access each one at a later time to make adjustments.



ImportMcMaster

This is custom import function specifically for STEP files which saves a couple of steps for the user.

1. First, it allows you to select a step file to import.
2. It creates a new layer with the name of the file you just imported and assigns the object to that layer.

3. It then creates a block instance with the imported geometry with the origin point being at the world origin (0,0,0)

This is intended to help us keep track of all the hardware used in the file.

MoveOrtho

This is a custom move function, much like the point to point standard move function that rhino has to offer, but it restricts the motion in one of the world orthogonal axes.

This allows the user to easily align faces by clicking two arbitrary points.

1. The user selects objects to move.
2. The user picks a direction to restrict movement within.
3. The user selects a start and end point to create the translation with.

LabelPart

This is a custom leader function that will populate the leader text with the object the leader tip is anchored to through the detail.

Work in progress...

Didn't seem to work as of a recent test... stay tuned...